

Assignment: Practical 1

Week 1 - From Python to C++

General requirements

- I. Add your name, surname, and student number as a comment line at the top of each of your programs;
- II. Use sensible comments to **explain** your program logic where necessary.
- III. When you submit to CodeGrade, make sure to submit all the files together; **SU1-1.cpp, SU1-2.cpp and SU1-3.cpp**

Prac 1-1 Calculating properties of a cone

You need to calculate the surface area, volume, and slant height of a cone with a given radius and height provided by the user.

Create a C++ program that asks the user to enter the radius (cm) and height (cm) of a cone. Calculate the surface area, volume, and slant height of the cone using applicable math functions. Display the calculated values with labels. Format the calculated values as follows: the area formatted to 4 decimals, the volume formatted to 2 decimals, and the slant height formatted to 3 decimals. Dimensions displayed as floating-point values with 1 decimal.

Rename your **main.cpp** to **SU1-1.cpp** and submit to CodeGrade along with the other programs.

Example run

```
Cone Calculator
Enter the radius (cm): 5.0
Enter the height (cm): 12.0

Surface area of the cone is 282.7433 square cm
Slant height of the cone is 13.000 cm
Volume of the cone is 314.16 cubic cm

**Cone dimensions:**
Radius: 5.0 cm
Height: 12.0 cm
```

Python reference code

```
import math

radius = float(input("Enter the radius (cm): "))
height = float(input("Enter the height (cm): "))

slant = math.sqrt(radius**2 + height**2)
area = math.pi * radius * (radius + slant)
volume = (math.pi * radius**2 * height) / 3

print(f"\nSurface area of the cone is {area:.4f} square cm")
print(f"Slant height of the cone is {slant:.3f} cm")
print(f"Volume of the cone is {volume:.2f} cubic cm")

print("\n**Cone dimensions:**")
print(f"Radius: {radius:.1f} cm")
print(f"Height: {height:.1f} cm")
```

Prac 1-2 ASCII and character conversion

You need to provide the ASCII code of a character or the character for a given ASCII value.

Create a C++ program that asks the user to enter a character. Determine and display its corresponding ASCII value. Ask the user to enter an ASCII value (integer between 0 and 127 - validation not required). Determine and display the corresponding character.

Rename your *main.cpp* to *SU1-2.cpp* and submit to CodeGrade along with the other programs.

Example run

ASCII Character Conversion Tool

Enter a character: A
The ASCII value is: 65

=====

Enter an ASCII value (0-127): 36
The character is: \$

Python reference code

```
print("ASCII Character Conversion")

char = input("Enter a character: ")
print(f"The ASCII value is: {ord(char)}")

print("\n=====")

code = int(input("Enter an ASCII value: "))
print(f"The character is: {chr(code)}")
```

Prac 1-3 Checking Eligibility for Driving License

You need to determine whether a user is eligible for a driving license based on age and residency status.

Create a C++ program that asks the user to input their age and whether they are a resident (Y/N). The program should check that the age is 16 or older and that residency status is 'Y' (Yes). If both conditions are met, print "You are eligible for a driver's license.". Otherwise, specify which condition(s) failed.

Rename your *main.cpp* to *SU1-3.cpp* and submit to CodeGrade along with the other programs.

Example run

```
Enter your age: 17
Are you a resident? (Y/N): Y

You are eligible for a driver's license.

Enter your age: 15
Are you a resident? (Y/N): Y

You are not old enough to apply.

Enter your age: 18
Are you a resident? (Y/N): N

You must be a resident to apply.
```

Python reference code

```
age = int(input("Enter your age: "))
resident = input("Are you a resident? (Y/N): ").upper()

if age >= 16:
    if resident == "Y":
        print("You are eligible for a driver's license.")
    else:
        print("You must be a resident to apply.")
else:
    print("You are not old enough to apply.")
```

Grading Criteria:

Item	Description	Marks
Practical 1-1 (Cone Calculator)		[15]
	1. Static analysis with clang-tidy (readability and good programming practices)	/3
	2. Program compiles and runs. Auto-test provides input and program's output is matched with expected output. (Syntax errors will result in zero marks for this item)	/4
	3. Import the math header file	/1
	4. Obtain radius and height values using appropriate data types (float/double)	/2
	5. Calculate surface area, volume, and slant height using formulas	/2
	6. Display values with correct decimal rounding and formatting	/3
Practical 1-2 (ASCII Conversion)		[13]
	1. Static analysis with clang-tidy (readability and good programming practices)	/3
	2. Program compiles and runs. Auto-test provides input and program's output is matched with expected output. (Syntax errors will result in zero marks for this item)	/4
	3. Obtain character input from the user	/1
	4. In a single statement, convert character to ASCII value using type casting and display it	/2
	5. Obtain ASCII value (integer input) and validate	/1
	6. In a single statement, convert ASCII value to character using type casting and display it	/2
Practical 1-3 (Driver's License Eligibility)		[13]
	1. Static analysis with clang-tidy (readability and good programming practices)	/3
	2. Program compiles and runs. Auto-test provides input and program's output is matched with expected output. (Syntax errors will result in zero marks for this item)	/3
	3. Obtain age (integer) and residency status (char) from the user	/2
	4. Resident input updated to uppercase character	/1
	5. Determine eligibility based on age and residency	/4